

TECHNICAL DATA

MQ-136 GAS SENSOR

FEATURES

Fast response and High sensitivity

Stable and long life

Simple drive circuit

APPLICATION

They are used in air quality control equipments for buildings/offices, are suitable for detecting of H₂S.

SPECIFICATIONS

A. Standard work condition

Symbol	Parameter name	Technical condition	Remarks
V _c	Circuit voltage	5V±0.1	AC OR DC
V _H	Heating voltage	5V±0.1	AC OR DC
R _L	Load resistance	can adjust	
R _H	Heater resistance	31Ω±5%	Room Tem
P _H	Heating consumption	less than 800mw	

B. Environment condition

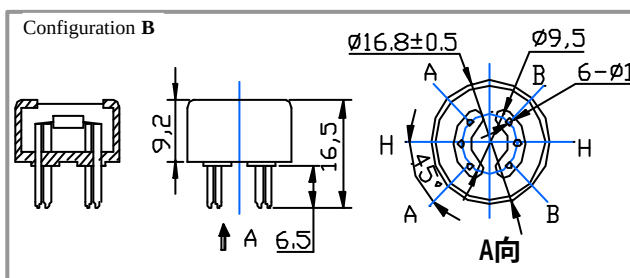
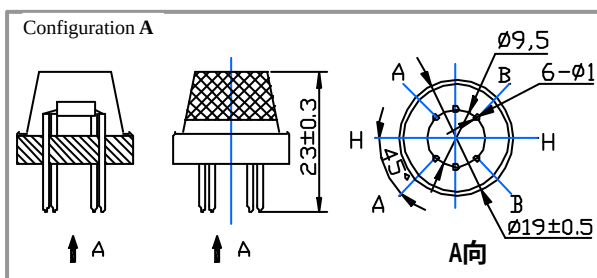
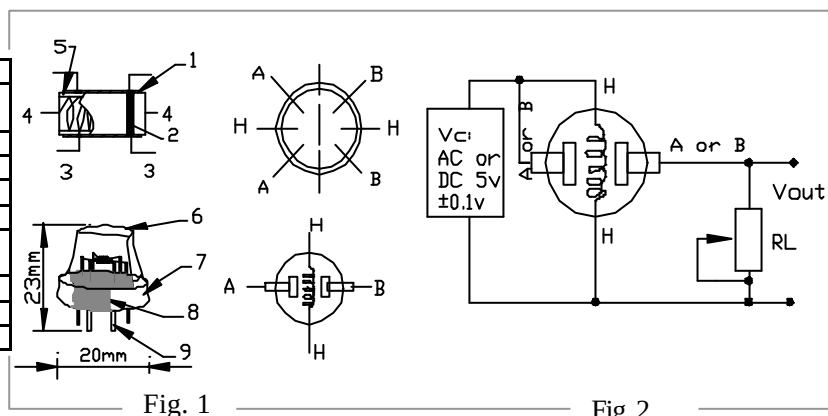
Symbol	Parameter name	Technical condition	Remarks
Tao	Using Tem	-10°C-45°C	minimum value is over 2%
Tas	Storage Tem	-20°C-70°C	
R _H	Related humidity	less than 95%Rh	
O ₂	Oxygen concentration	21%(standard condition)Oxygen concentration can affect sensitivity	

C. Sensitivity characteristic

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Symbol	Parameter name	Technical parameter	Remark 2
Rs	Sensing Resistance	30K Ω -200K Ω (10ppm H ₂ S)	Detecting concentration scope : 1-100ppm H ₂ S
α (20/5) H ₂ S	Concentration Slope rate	≤ 0.65	
Standard Detecting Condition	Temp: 20°C $\pm$ 2°C Vc:5V $\pm$ 0.1 Humidity: 65% $\pm$ 5% Vh: 5V $\pm$ 0.1		
Preheat time	Over 24 hour		

D. Structure and configuration, basic measuring circuit

Parts	Materials
1 Gas sensing layer	SnO ₂
2 Electrode	Au
3 Electrode line	Pt
4 Heater coil	Ni-Cr alloy
5 Tubular ceramic	Al ₂ O ₃
6 Anti-explosion network	Stainless steel gauze (SUS316 100-mesh)
7 Clamp ring	Copper plating Ni
8 Resin base	Bakelite
9 Tube Pin	Copper plating Ni



Structure and configuration of MQ-136 gas sensor is shown as Fig. 1 (Configuration A or B), sensor composed by micro Al₂O₃ ceramic tube, Tin Dioxide (SnO₂) sensitive layer, measuring electrode and heater are fixed into a crust made by plastic and stainless steel net. The heater provides necessary work conditions for work of

sensitive components. The enveloped MQ-136 have 6 pin ,4 of them are used to fetch signals, and other 2 are used for providing heating current.

Electric parameter measurement circuit is shown as Fig.2

E. Sensitivity characteristic curve

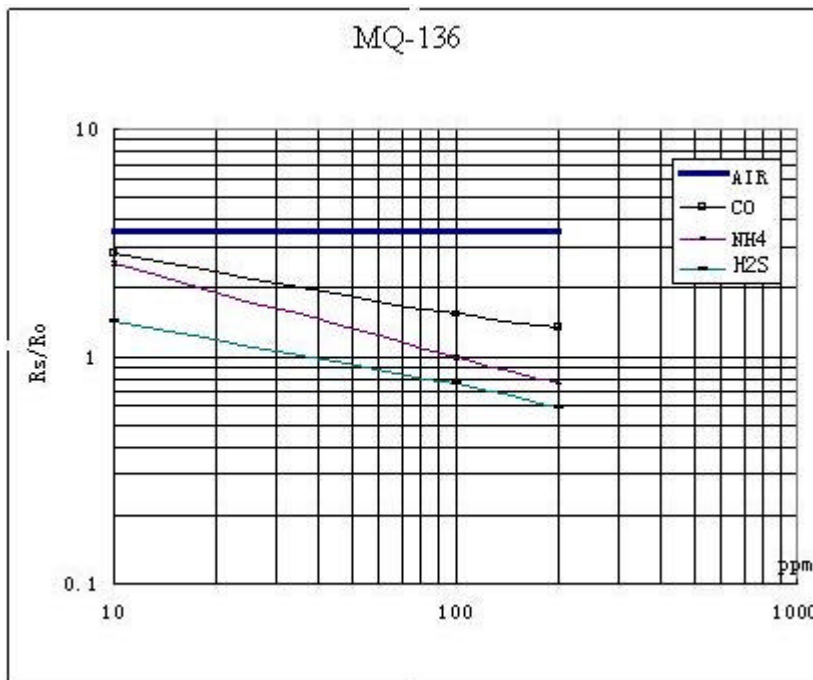


Fig.3 is shows the typical sensitivity characteristics of the MQ-136 for several gases.

in their: Temp: 20°C、

Humidity: 65%、

O₂ concentration 21%

RL=20kΩ

Ro: sensor resistance at 10ppm of H₂S in the clean air.

Rs: sensor resistance at various concentrations of gases.

Fig.3 sensitivity characteristics of the MQ-136

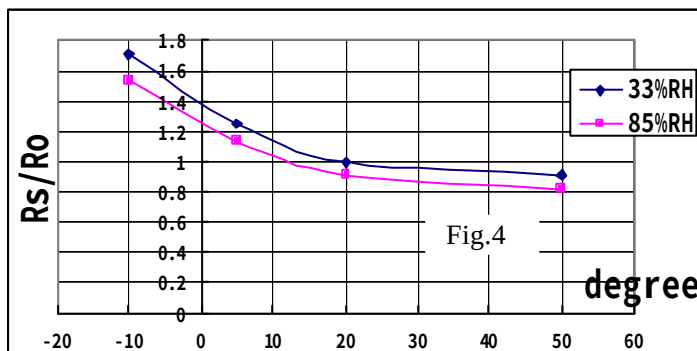


Fig.4 is shows the typical dependence of the MQ-136 on temperature and humidity.

Ro: sensor resistance at 10ppm of H₂S at 33%RH and 20 degree.

Rs: sensor resistance at 10ppm of H₂S at different temperatures and humidity.

SENSITIVITY ADJUSTMENT

Resistance value of MQ-136 is difference to various kinds and various concentration gases. So, When using this components, sensitivity adjustment is very necessary. we recommend that you calibrate the detector for 10ppm H₂S concentration in air and use value of Load resistance that(R_L) about 20 KΩ(10KΩ to 47 KΩ).

When accurately measuring, the proper alarm point for the gas detector should be determined after considering the temperature and humidity influence.

